

CSA 0711

39

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I T. Subaham (192511121) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Subaham

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5
Problem Understanding	20%	2	2	2	2	1
Method Selection	20%	2	1	1	1	1
Solution Process	25%	1	2	1	1	2
Accuracy of Calculation	20%	2	2	2	2	1
Interpretation of Results	15%	1	1	2	1	2

8 38

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Total Marks 38 /40

Signature of the Faculty

Subnetting the Network :

Network Address : 192.168.20.0/24

Lab A $\rightarrow$ /26

Lab B $\rightarrow$ /27

Lab C $\rightarrow$ /28

Calculate Subnets :

Lab A (/26) :

i) Subnet Mask : 255.255.255.192

ii) Block Size : 64

iii) Network Address : 192.168.20.0

iv) Valid Host Range = 192.168.20.1 - 192.168.20.62

v) Broadcast Address = 192.168.20.63

vi) Total IPs = 64

vii) Usable hosts = 62

Lab B (/27) :

Next available subnet starts at 192.168.20.64

i) Subnet Mask = 255.255.255.224

ii) Block Size = 32

iii) Network Address = 192.168.20.64

iv) Valid Host Range = 192.168.20.65 - 192.168.20.94

v) Broadcast Address = 192.168.20.95

vi) Total IPs = 32

vii) Usable Hosts = 30

min Department (128):

Next Subnet:

1. Network = $172.16.0.224$

2. Host Range = $172.16.0.225 - 172.16.0.238$

3. Broadcast = $172.16.0.239$

4. Usable Hosts = 14

The departments receive subnet sizes based on their host requirements. The remaining IP address space is preserved for future scalability, ensuring efficient address utilization.

4. Subnetting $172.16.10.0/24$ Using 127 :

Network = $172.16.10.0/24$

Subnet Mask = 127

Number of Subnets

Borrowed bits = 3

Number of Subnets: $2^3 = 8$

Hosts Per Subnet

Host bits = 5

Total IPs = $2^5 = 32$

Usable Hosts = $32 - 2 = 30$